



BEST PRACTICE SERIES

Utilization of Existing Onboard Electronics

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
				Utilization of Existing Onboard Electronics.....0
				1.0 Introduction.....1
				2.0 Best Practice Description.....1
				3.0 Implementation Steps.....6
				4.0 Benefits.....7
				5.0 Resources Required.....8
				6.0 Supporting Attachments / References.....8
				7.0 Related Best Practices.....8
				8.0 Acknowledgements.....8

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1.0 Introduction

Electronic Equipment Data is one of the five key Condition Monitoring elements that enable the Maintenance and Repair goal of scheduling 80% of maintenance and allowing no more than 20% of maintenance to be unscheduled. Value to the customers can be achieved by establishing foundational skills and experience with onboard electronics on existing machines. When a customer's population of Cat machines with Electronic Control Modules (ECM) reaches a critical mass, each customer site could justify the investment in time and resources to capture information from those machines. This Best Practice recommends customer managers already have the skills needed to identify instances or trends that require further attention. This recommendation is also an initial step for customers who are acquiring and utilizing more advanced technologies.

Product Health is understood as the use of technology to reduce downtime. After an alarm or failure, technology assists in the product health activity of troubleshooting. However, a potentially more significant benefit is the use of technology to pinpoint individual events and/or trends in order to proactively perform maintenance and repairs. Planned service is far less disruptive to production and is normally less expensive than post-failure repairs.

Operational data is also collected by the onboard electronics systems on many machines. At the very least, idle hours and fuel consumption are indicators of the work done by each individual machine. Machines equipped with more advanced technologies give insights to payloads, cycle times and application severity.

Achieving the lowest cost per ton/m³/yd³ with Cat machines involves optimization of payload, cycle time and availability. This Best Practice gives equipment managers a chance to listen to what their existing Cat machines can tell them as they manage all three elements.

2.0 Best Practice Description

2.1 Basis of the Best Practice

Several policy decisions are required in order for the Caterpillar equipment owner to realize the potential value of this Best Practice. The most crucial policy decision asserts that the Product Status Report (PSR) generated by Cat ET contains actionable insights. Additionally, two related policy decisions are 1) that the PSR should be downloaded and communicated to the customer's equipment manager 2) as part of their regular Preventive Maintenance (PM) schedule. This Best Practice had its origins in a Continuous Improvement (CI) project chartered to establish the practicality of these policy decisions.

At a single customer location, the CI project generated PSRs from 15 machines averaging 4.5 years old and 26,000 service meter hours. Actionable items included four backlog items for product health reasons and two backlog items to re-configure ECM settings to improve productivity. The PSRs identified pre-failure repair opportunities that had not been caught as part of the customer's inspection and PM activities. In addition, one operator training opportunity was

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265

identified and one operational issue was unresolved pending further investigation by the customer. These operational improvements promise to decrease machine cycle times. The CI project demonstrated the value of downloading the PSR.

Regular review of ECM data is practical. At the site of the CI project, the equipment manager would expect about one PSR in the e-mail inbox each day. Scanning one PSR per day would take just moments. When a PSR identifies an actionable event, the common courses of action would be to create a backlog item or to ask the operations supervisor to review the circumstances.

This Best Practice concludes that Cat ET downloads can and should be performed by customer PM technicians. At the site of the CI project, a veteran PM technician with minimal PC experience performed all 15 downloads in less than one shift. With only the assistance of the job-aid, the PM technician was able to perform the process after the first couple of downloads. It is reasonable to expect PM technicians to create and forward Product Status Reports. While troubleshooting failures using Cat ET does require experienced, trained technicians; capturing ECM data is a far less complex task.

Once the Best Practice is put into place, the process is short and straightforward. At specified PM intervals, the PM technician connects to the machine with the appropriate Cat ET cable, chooses to create the Product Status Report, prints the report to a .pdf file, and e-mails the file to the equipment manager and Cat dealer. When the equipment manager receives the report, scanning the diagnostic and event codes identifies any conditions that would either create a backlog item or would be passed to the operations management for review. Depending on the customer's priorities, the equipment manager may also enter readings into a spreadsheet from the current totals section of the Product Status Report for one or more ECMs. This spreadsheet would provide trending for operational information like fuel consumption or percentage idle time. When the Cat dealer receives the report, they will summarize reports from all machines at the site in preparation for review at the quarterly communications meetings with the customer site management.

2.2 Scheduling the Product Status Report

For customers that perform pre-PM inspections, this task is appropriate within that checklist and offers the added benefit of creating backlog items that may be resolved during the downtime allocated to the PM. If pre-PM inspections are not performed, creating and e-mailing this report as part of the PM checklist is recommended.

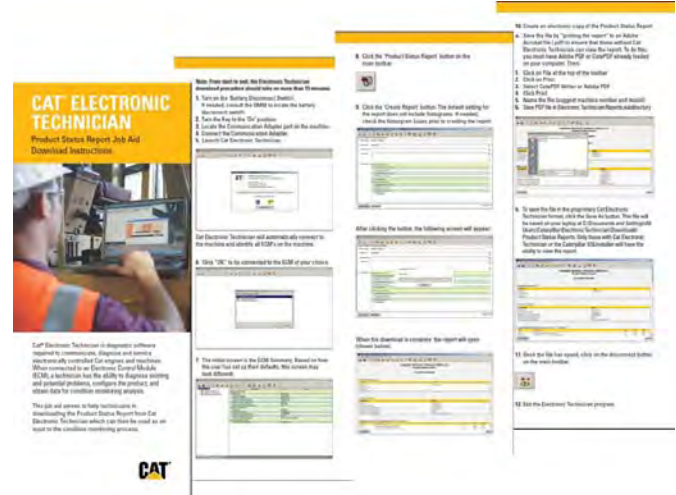
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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265

2.3 Job-aid for the Creating the Product Status Report

In the supporting attachments for this Best Practice is a document developed to help PM technicians gain confidence with the laptop computer and the Cat ET software. The media number for ordering this document is PEDJ0122.

The document includes step-by-step instructions and including Cat ET screens.

The document is brief and is intended to be used by the technician during the PM service.



2.4 Equipment Manager Scan of PSRs

Upon receiving the e-mail containing the Product Status Report, the customer's equipment manager reviews the codes and events for each ECM that exists on this particular machine. Note that when hour meter readings are displayed in these sections, they are in terms of the individual ECM's diagnostic clock and may vary from the machine's hour meter reading.

The Logged Diagnostic Codes section tracks issues where the ECM is reporting possible faulty sensors or input to the ECM. Backlog items to address these diagnostics are recommended.

The Logged Event Codes section tracks any number of issues where the component/system managed by the ECM is performing out of specifications. Experienced equipment managers make the decision to whether to create a backlog item or to notify operations.

For any one diagnostic code, the PSR displays the diagnostic clock reading at the time of both the first and most recent occurrence. The ECM allocates memory to store up to 127 or 255 occurrences of any one diagnostic code. The other stored occurrences are available to the technician who would perform repair or troubleshooting services.

The Active Diagnostic Codes section and the Active Event Code section display the same information as the logged codes sections for issues that were present when the Product Status Report was generated. The equipment manager's actions for active codes follow the same guidelines as for logged codes.

Logged Diagnostic Codes [Diagnostic Clock = xx,xxx hours]				
Code	Description	Occ.	First	Last
No Logged Diagnostic Codes				
Logged Event Codes [Diagnostic Clock = xx,xxx hours]				
Code	Description	Occ.	First	Last
No Logged Events				
Active Diagnostic Codes				
Code	Description			
No Active Codes				
Active Event Codes				
Code	Description			
No Active Events				

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Utilization of Existing Onboard Electronics

DATE
1/12/2015

CHG
NO
00

NUMBER
1112-1.02-1265

2.5 Equipment Manager Trending Process

The Product Status Report also contains a section labeled Current Totals for each ECM. The contents of this section vary for each component/system managed by an ECM. The example shown is for an engine ECM on a Wheel Loader.

Current Totals		
Description	Value	Unit
Total Fuel	10911	Gal
Total Idle Time	738:57	hours
Total Idle Fuel	2817	Gal
Total Max Fuel	24476	Gal
Total Operating Hours	1393	hours
Engine Starts	2344	
Lifetime Total Engine Revolutions	102930984	rev
Average RPM	1232.41	RPM
Percentage Idle Time	53.02	%
Average Fuel Rate	7.84	GPH
Overall Load Factor	44	%

The values displayed are for the life of the ECM. For any values that the customer determines a trend would be beneficial, it is necessary to develop a process for recording the information into a separate database each time a Product Status Report is generated. Recommended trending of data from the engine ECM would include Total Fuel, Total Idle Time and Total Idle Fuel.

A mock up of spreadsheet is shown. Note, all of the data is hypothetical and is not intended to serve any purpose other than to suggest what an analysis of trends might include. The color coding shows what must be manually recorded from each Product Status Report versus the labels and formulas that can provide meaning to the analysis.

Unit #:	L7654	Current Totals			Trend		
	Engine Diagnostic Clock	Fuel (Gal)	Idle Time	Idle Fuel	Fuel Consumption (Gal per hour)	% Idle Time	% Idle Fuel
Date							
10/1/2010	143	922	112	305	6.5	78.6%	33.1%
11/1/2010	401	2,909	255	903	7.7	55.1%	30.1%
12/1/2010	641	4,745	389	1,463	7.7	55.9%	30.5%
1/1/2011	898	6,775	520	1,965	7.9	51.2%	24.7%
2/1/2011	1,137	8,735	630	2,404	8.2	46.1%	22.4%
3/1/2011	1,393	10,911	738	2,817	8.5	42.0%	19.0%

The anticipated value of this type of trending is to monitor the operations at the customer's site. The equipment manager's determination of what should be trended is limited only by what is available from the ECMs on the Cat machines in their fleet.

2.6 Cat Dealer Summary

Forwarding the Product Status Reports to the dealer enables the dealer to generate a high level overview for the customer. Cat Global Mining encourages quarterly communications meetings involving customer and dealer leadership to review the all aspects of their relationship, including a review of product and product support related issues. The dealer's review of the PSR can leverage their experience with similar products, similar applications and or leverage their access to Caterpillar for solutions and support.

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265

Another Best Practice (Communicating VIMS Events Using Critical Event Advice) recommends a form for summarizing insights from onboard electronics. The benefit of this communication tool is that it provides a higher level view of the more significant issues that are identified by the Product Status Reports.

What this level of summary contributes is the ability to determine if an issue needs attention beyond the one-at-a-time backlog management approach, since it is anticipated that the customer's equipment manager took action on the individual codes and events as part of their daily review of the PSRs.

These communication meetings help identify those circumstances where closer partnering between the customer and the dealer have the ability to lower the cost per ton/m³/yd³. The Critical Event Advice, or something comparable, is one technique available to improve those meetings.

2.7 Metrics and Ongoing Process

This best practice does not recommend the creation of a new metric regarding the creation of the PSR. We assume that standard operating procedure utilizes a checklist for each PM service interval and that this procedure has the capability of adding the Cat ET Product Status Report (PSR) to the checklists. That assumption includes a metric that tracks the completion of the PM services and therefore tracks the creation of the PSR.

Backlog items created by the review of the report are also assumed to be tracked by the customer as part of the normal backlog process. A general category of Electronic Equipment Data would be recommended if the backlog system tracks the source of the backlog item.

Operational improvements are likely to occur as a result of the review of the reports. The trending spreadsheet suggested in this best practice can be used as both an analysis tool as well as a tracking mechanism.

Including the Critical Event Advice in the quarterly meetings is a new source of insight into the customer and dealer relationship, but is not what would be considered a metric.

Over time, the advances in technology available to Cat machine owners are expected to surpass the actionable insights that this Best Practice recommends. However the substantial population of machines in use today with this level of technology suggests that Cat ET Product Status


Dealer Name

Critical Event Advice

Sample Periodic Summary Report
Sample Site

Critical Event Advice				
Unit #	Event	Qty	Reported	Diagnostic Clock

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265

Reports will be a beneficial component of a comprehensive Condition Monitoring process for quite some time.

3.0 Implementation Steps

3.1 PM Technician

A laptop computer for dedicated use by the PM technicians is recommended. The customer can contact the Cat dealer to acquire and authorize the Cat ET software. A communications adapter cable is necessary to connect the PM technician's laptop computer to the Cat machines.

The recommended process is to create the Product Status Report on the PM technician's laptop; to print the report to a file in .pdf format; and to e-mail the report to the specified customer and dealer personnel. Software is required to create the .pdf file format. At the CI project, the software CutePDF™ from ACRO Software Inc. was used.

The recommended process is to store the .pdf format reports in a folder that is convenient for the PM technician to access from the e-mail software. When the PM technician's laptop computer is configured, this folder should be created.

Information for all of these products is included in the Resources Required section of this Best Practice.

Customer information technology policies will dictate the e-mail product for use on the PM technician's laptop computer.

3.2 Equipment Manager

The equipment manager can perform the reviews of the Product Status Report using commonly configured laptop or desktop PCs. Customer information technology policies will dictate the e-mail product for use on the equipment manager's computer as well as the data analysis software used for creating the trending spreadsheet. At the customer site where the CI project occurred, common Microsoft Office products were used.

As an alternative to the PM technician creating a .pdf copy of the Product Status Report, Cat ET creates a file in proprietary Caterpillar file format every time the report is generated. This file is viewable on any computer with Cat ET installed or can be viewed with a Caterpillar supplied add-in. This add-in, distributed as XSLInstaller.exe, is available to dealers on the Caterpillar Intranet. Cat ET documentation will identify where this proprietary file is stored on the PM technician's PC.

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265
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3.3 Initializing the Cat Machines

This Best Practice recommends that an experienced technician connect to each Cat machine to initialize the header information displayed in the Cat ET Product Status Report.

The recommendation is that the Product ID parameter contains the serial number for the Caterpillar Machine. The Equipment ID parameter is most useful when it contains both the customer's assigned unit number and the model of the machine. When this set up step is taken, knowing which machine generated this report does not rely on the PM technician entering accurate information when naming the .pdf file.

Caterpillar Electronic Technician 2009B v1.0 Product Status Report	
1/28/2011 9:39 PM	
Product Status Report	
Parameter	Value
Product ID	JMSyyyyy
Equipment ID	
Comments	

The team that developed this Best Practice considered, and rejected, clearing of all of the event codes in on the ECMs during this initialization process. The team recommends that the customer establish a policy to clear the codes after the cause of the code was identified and corrected.

3.4 Training

This best practice recommends training for the PM technicians that focuses on the creation of the Product Status Report. Using the job-aid referenced below, hands-on training by an experienced technician should be effective in most situations. Even though the Resources Required section of this Best Practice includes a course number for technician training with Cat ET, that training material covers much more than is required for the task at hand.

4.0 Benefits

This Best Practice is one way to include Electronic Equipment Data in the customer's Condition Monitoring efforts. The technology required to monitor the machine's components and systems is already on the machines and has been recording data since the machine was new. The incremental cost in time, software and hardware is relatively small, compared to the potential.

As with all elements of Condition Monitoring, this investment only lowers cost per ton/m³/yd³ when the information is acted upon. Disciplined execution of a backlog management process can turn information from onboard electronics into higher machine availability, lower operating costs and operational improvements.

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265

5.0 Resources Required

NEXG5007 Customer All Product Cat ET
 or
 NEXG5076 Dealer All Product Cat ET (For Customer use)
 317-7484 Communications Adapter 3 Group
 28657 SCORM DLMS Web based training - Caterpillar Electronic Technician Training

Creating the .pdf version of the Cat ET Product Status Report requires software on the PM technician's laptop computer. Caterpillar uses CutePDF™ from ACRO Software Inc. This product and the required converter can be downloaded from:

<http://www.cutepdf.com/products/cutepdf/writer.asp>

6.0 Supporting Attachments / References

PEDJ0122 Caterpillar Electronic Technician –
 Product Status Report Job Aid Download Instructions

7.0 Related Best Practices

0608-1.1-1124 – Communicating VIMS Events Using Critical Event Advice

8.0 Acknowledgements

Support for the Continuous Improvement project that led to the creation of this best practice came from:

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Utilization of Existing Onboard Electronics	DATE 1/12/2015	CHG NO 00	NUMBER 1112-1.02-1265
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